Unit 06

Scientific Research

1. DESIGNING EXPERIMENTS

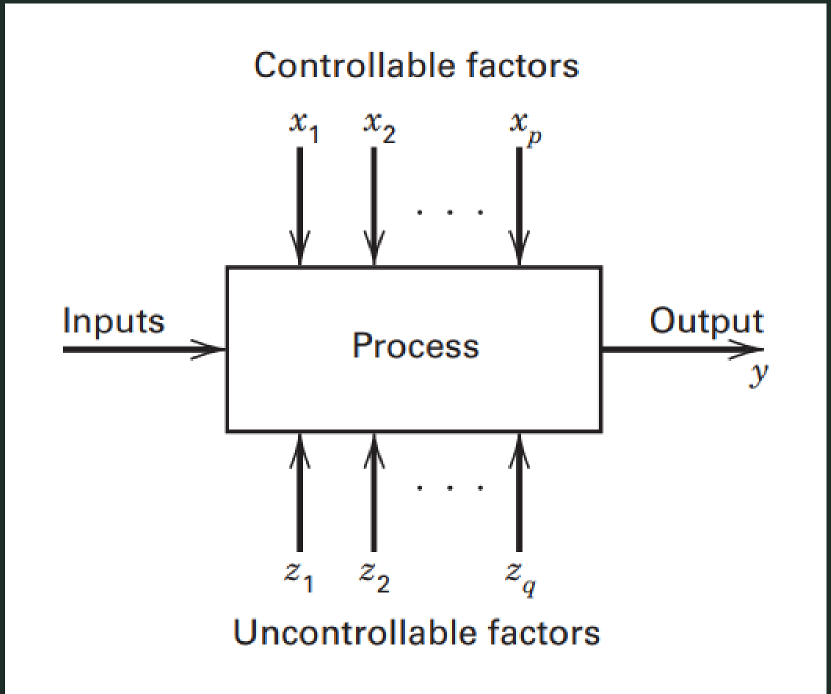
EXPERIMENTS
Definition of experiment

- *An **experiment** is a test or series of runs in which purposeful changes are made to the input variables of a process or system so that we may observe and identify the reasons for changes that may be observed in the output response.*
- Observations on a system or process can lead to theories or hypotheses about what makes the system work, but experiments of the type described above are required to demonstrate that these theories are correct.
- We may want to:
  - determine which input variables are responsible for the observed changes in the response,
  - develop a model relating the response to the important input variables,
  - and use this model for process or system improvement or other decision-making.

EXPERIMENTS

Model of a process or system

- We can usually visualize the process as a combination of operations, machines, methods, people, and other resources that transforms some input (often a material) into an output that has one or more observable **response variables**.
- Some of the process variables and material properties $x_1, x_2, \ldots, x_p$ are **controllable**, whereas other variables such as environmental factors or some material properties $z_1, z_2, \ldots, z_q$ are **uncontrollable.**



EXPERIMENTS

Example

- Consider an experiment in which we want to observe the effect on the system's response time of varying the size of the main memory available in the system and the degree of multiprogramming allowed. (The degree of multiprogramming is the number of application that are allowed to time-share the processor)
- *Controllable factors*: degree of multiprogramming (A) and memory size (B)
- *Response variable*: the system's response time (in seconds)

	B (Mbytes)		
A	32	64	128
1	0.25	0.21	0.15
2	0.52	0.45	0.36
3	0.81	0.66	0.50
4	1.50	1.45	0.70

EXPERIMENTS
Objectives of Experiments

- The objectives of the experiment may include the following:
  - Determining which variables are most influential on the response variable *y*
  - Determining where to set the influential *x*'s so that *y* is almost always near the desired nominal value
  - Determining where to set the influential *x*'s so that variability in *y* is small
  - Determining where to set the influential *x*'s so that the effects of the uncontrollable variables $z_1, z_2, \ldots, z_q$ are minimized.

EXPERIMENTS
Terminology

- **Response variable**
  - Measured output value (E.g. total execution time)
- **Factors**
  - Input variables that can be changed (E.g. cache size, clock rate, bytes transmitted)
- **Levels**
  - Specific values of factors - Continuous (~bytes) or discrete (type of system)
- **Replication**
  - Completely re-run experiment with same input levels
  - Used to determine impact of measurement error
- **Interaction**
  - Effect of one input factor depends on level of another input factor

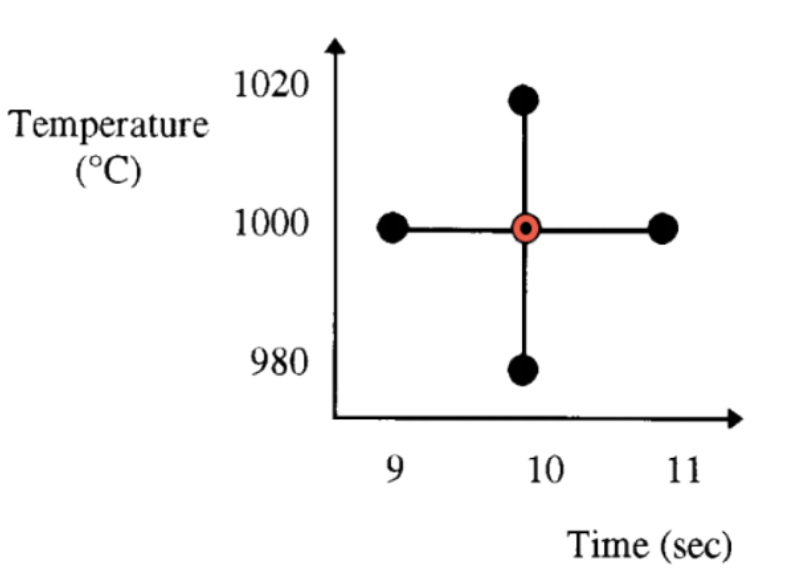
EXPERIMENTATION STRATEGIES
Educated guessing

- ***Educated guessing*** *strategy:*
  1. *Select arbitrary combination of levels for the factors;*
  2. *Test and observe behavior;*
  3. *Change one or two factors at a time, then re-test;*
- Widely used. It often works reasonably well, too, because the experimenters often have a great deal of technical or theoretical knowledge of the system they are studying, as well as considerable practical experience.
- Limitations:
  - Suppose the initial best-guess does not produce the desired results. Now the experimenter has to take another guess at the correct combination of factor levels. This could continue for a long time, without any guarantee of success.
  - Suppose the initial best-guess produces an acceptable result. Now the experimenter is tempted to stop testing, although there is no guarantee that the *best* solution has been found.

EXPERIMENTATION STRATEGIES
One-factor-at-a-time (OFAT)

- **One-factor-at-a-time** approach:
  1. Selecting a starting point, or **baseline** set of levels, for each factor,
  2. Successively varying each factor over its range with the other factors held constant at the baseline level.

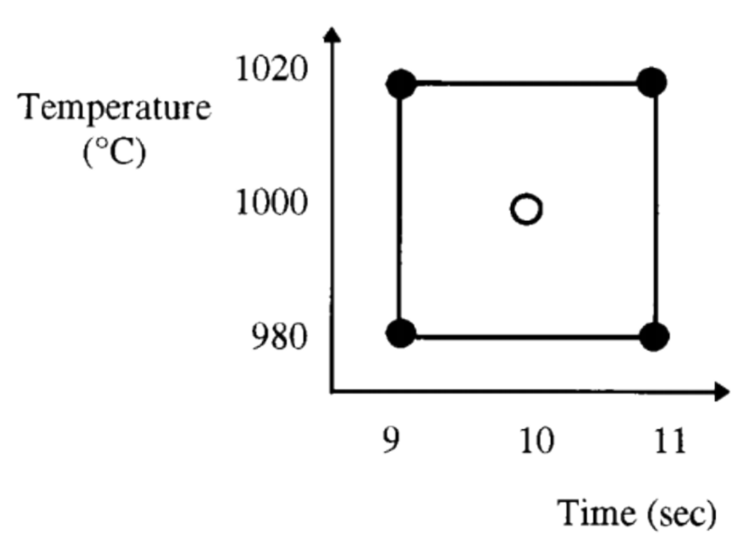
- Also widely used;
- Can achieve good results as long as there are no **interaction** effects; An interaction is the failure of one factor to produce the same effect on the response at different levels of another factor.



EXPERIMENTATION STRATEGIES
Factorial

- **Factorial** approach:
  1. Select levels for each factor;
  2. Vary the factors simultaneously, in a systematic way;

- Estimation of main effects and interactions;
- Greater precision in the effect estimates;
- More efficient use of the experimental data;



EXPERIMENTATION STRATEGIES
Fractional Factorial

- Generally, if there are *k* factors, each at two levels, the factorial design would require 2^k runs.
- Clearly, as the number of factors of interest increases, the number of runs required increases rapidly;
- This quickly becomes infeasible from a time and resource viewpoint.
- A **fractional factorial experiment** is a variation of the basic factorial design in which only a subset of the runs is used.
- Fractional factorial designs are used extensively in industrial research and development, and for process improvement.

EXPERIMENTAL DESIGN

- Experimental design methods have found broad application in many disciplines.
- We may view experimentation as part of the scientific process and as one of the ways by which we learn about how systems or processes work.
- Generally, we learn through a series of activities in which:
  - We make conjectures about a process,
  - <u>*Perform experiments to generate data from the process*</u>,
  - and then use the information from the experiment to establish new conjectures, which lead to new experiments, and so on.

EXPERIMENTAL DESIGN
Example

EXAMPLE 1.6 **Designing a Web Page**

A lot of business today is conducted via the World Wide Web. Consequently, the design of a business' web page has potentially important economic impact. Suppose that the website has the following components: (1) a photoflash image, (2) a main headline, (3) a subheadline, (4) a main text copy, (5) a main image on the right side, (6) a background design, and (7) a footer. We are interested in finding the factors that influence the click-through rate; that is, the number of visitors who click through into the site divided by the total number of visitors to the site. Proper selection of the important factors can lead to an optimal web page design. Suppose that there are four choices for the photoflash image, eight choices for the main headline, six choices for the subheadline, five choices for the main text copy, four choices for the main image, three choices for the background design, and seven choices for the footer. If we use a factorial design, web pages for all possible combinations of these factor levels must be constructed and tested. This is a total of $4 \times 8 \times 6 \times 5 \times 4 \times 3 \times 7 = 80{,}640$ web pages. Obviously, it is not feasible to design and test this many combinations of web pages, so a complete factorial experiment cannot be considered. However, a fractional factorial experiment that uses a small number of the possible web page designs would likely be successful. This experiment would require a fractional factorial where the factors have different numbers of levels. We will discuss how to construct these designs in Chapter 9.

STATISTICAL DESIGN OF EXPERIMENTS

- **Statistical design of experiments** refers to the process of planning the experiment so that appropriate data will be collected and analyzed by statistical methods, resulting in valid and objective conclusions.
- Applicable to systems and processes subject to noise, experimental errors, uncertainties, etc.
- Necessary for the conclusions to have a quantifiable meaning;
- Helpful in avoiding errors due to personal biases or other artifacts of experimentation and analysis.
- There are two aspects to any experimental problem: the design of the experiment and the statistical analysis of the data. These two subjects are closely related because the method of analysis depends directly on the design employed.

BASIC PRINCIPLES
Of Statistical Design of Experiments

1. Randomization

- By randomization we mean that both the allocation of the experimental material and the order in which the individual runs of the experiment are to be performed are randomly determined.
- Statistical methods require that the observations (or errors) be independently distributed random variables. Randomization usually makes this assumption valid.
- By properly randomizing the experiment, we also assist in “averaging out” the effects of extraneous factors that may be present.

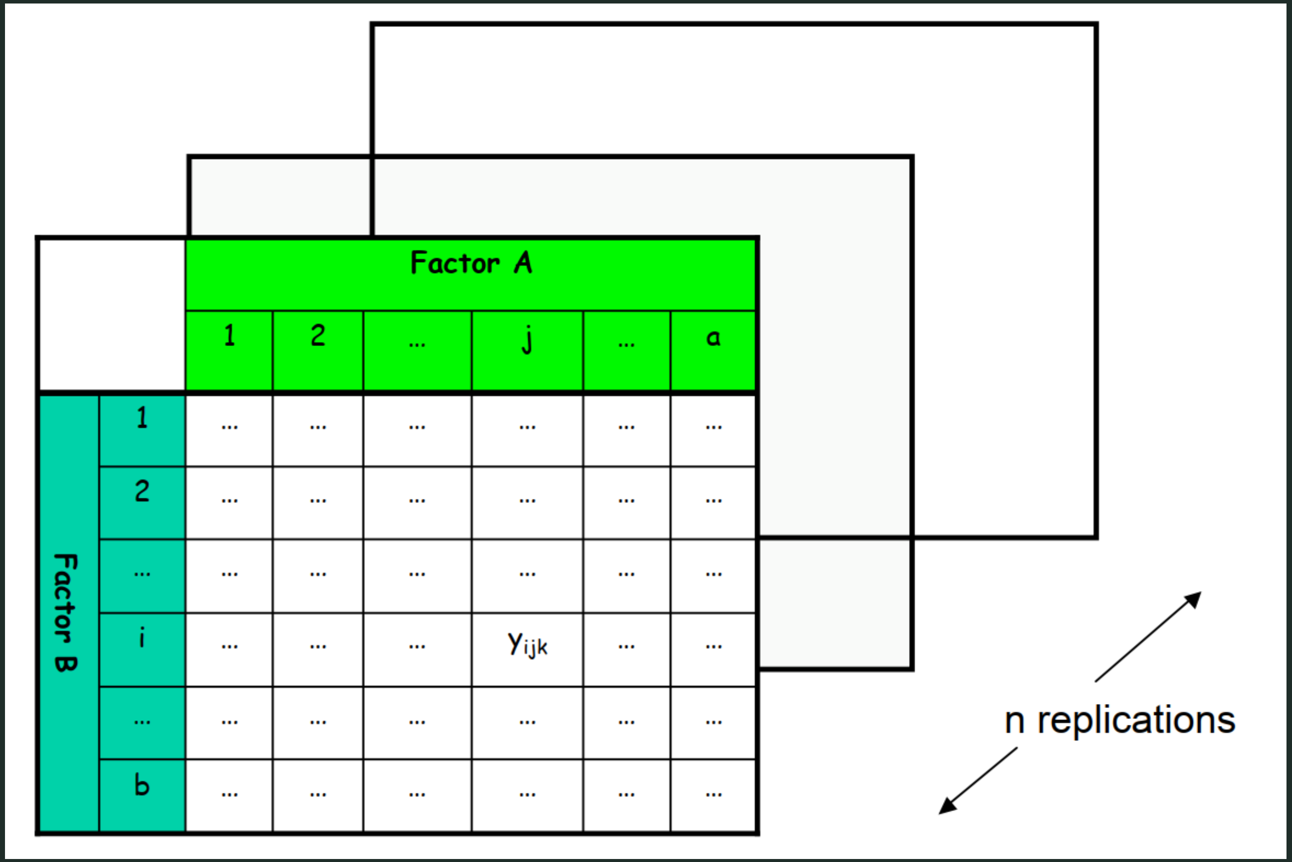
2. Replication

- By **replication** we mean an independent **repeat run** of each factor combination.
- Replication has two important properties.
  - First, it allows the experimenter to obtain an estimate of the **experimental error**. This estimate of error becomes a basic unit of measurement for determining whether observed differences in the data are really *statistically* different.
  - Second, if the sample mean is used to estimate the true mean response for one of the factor levels in the experiment, replication permits the experimenter to obtain a more precise estimate of this parameter.

BASIC PRINCIPLES
Of Statistical Design of Experiments

2. Replication

- Two Factors, n Replications



BASIC PRINCIPLES

Of Statistical Design of Experiments

3. **Blocking**

- Blocking is a design technique used to improve the precision with which comparisons among the factors of interest are made.
- Often blocking is used to reduce or eliminate the variability transmitted from **nuisance factors**—that is, factors that may influence the experimental response but in which we are not directly interested.
- Generally, a block is a set of relatively homogeneous experimental conditions.
- Typically, each level of the nuisance factor becomes a block. Then the experimenter divides the observations from the statistical design into groups that are run in each block.

GUIDELINES FOR DESIGNING EXPERIMENTS

The Recommended Procedure

1. Pre-experimental Planning
    a. Recognition of and statement of the problem
    b. Selection of the response variable
    c. Choice of factors, levels, and ranges
2. Choice of experimental design
3. Performing the experiment
4. Statistical analysis of the data
5. Conclusions and recommendations

GUIDELINES FOR DESIGNING EXPERIMENTS

Recognition of and statement of the problem

- This may seem to be a rather obvious point, but in practice often neither is it simple to realize that a problem requiring experimentation exists, nor is it simple to develop a clear and generally accepted statement of this problem.
- It is usually helpful to prepare a list of specific problems or questions that are to be addressed by the experiment.
- There are several broad reasons for running experiments and each type of experiment will generate its own list of specific questions that need to be addressed.

GUIDELINES FOR DESIGNING EXPERIMENTS

Recognition of and statement of the problem

- Some (but by no means all) of the reasons for running experiments include:
  - ***Factor screening or characterization.***
  - ***Optimization.*** After the system has been characterized and we are reasonably certain that the important factors have been identified, the next objective is usually optimization, that is, find the settings or levels of the important factors that result in desirable values of the response.
  - ***Confirmation.*** In a confirmation experiment, the experimenter is usually trying to verify that the system operates or behaves in a manner that is consistent with some theory or past experience.
  - ***Discovery.*** In discovery experiments, the experimenters are usually trying to determine what happens when we explore new materials, or new factors, or new ranges for factors.
  - ***Robustness.*** These experiments often address questions such as under what conditions do the response variables of interest seriously degrade? Or what conditions would lead to unacceptable variability in the response variables? …

GUIDELINES FOR DESIGNING EXPERIMENTS

Selection of the response variable

- In selecting the response variable, the experimenter should be certain that this variable really provides useful information about the process under study.
- Most often, the average or standard deviation (or both) of the measured characteristic will be the response variable.
- The experimenters must decide how each response will be measured, and address issues such as how will any measurement system be calibrated and how this calibration will be maintained during the experiment.
- The gauge or measurement system capability (or measurement error) is also an important factor. If gauge capability is inadequate, only relatively large factor effects will be detected by the experiment or perhaps additional replication will be required.

GUIDELINES FOR DESIGNING EXPERIMENTS

Choice of factors, levels, and ranges

- When considering the factors that may influence the performance of a process or system, the experimenter usually discovers that these factors can be classified as either **design factors** or **nuisance factors**.
- The design factors are those factors that the experimenter may wish to vary in the experiment and are the factors selected for study in the experiment.
- Nuisance factors may have large effects that must be accounted for, yet we may not be interested in them in the context of the present experiment.
  - Nuisance factors are often classified as **controllable, uncontrollable**, or **noise factors**.
  - A controllable nuisance factor is one whose levels may be set by the experimenter. The blocking principle, discussed in the previous section, is often useful in dealing with controllable nuisance factors.
  - If a nuisance factor is uncontrollable in the experiment, but it can be measured, an analysis procedure called the **analysis of covariance** can often be used to compensate for its effect.
  - When a factor that varies naturally and uncontrollably in the process can be controlled for purposes of an experiment, we often call it a noise factor. In such situations, our objective is usually to find the settings of the controllable design factors that minimize the variability transmitted from the noise factors.

GUIDELINES FOR DESIGNING EXPERIMENTS

Choice of factors, levels, and ranges

- Once the experimenter has selected the design factors, he or she must choose the ranges over which these factors will be varied and the specific levels at which runs will be made.
- The experimenter will also have to decide on a region of interest for each variable (that is, the range over which each factor will be varied) and on how many levels of each variable to use.
- **Process knowledge** is required to do this.
  - This process knowledge is usually a combination of practical experience and theoretical understanding.
  - It is important to investigate all factors that may be of importance and to be not overly influenced by past experience, particularly when we are in the early stages of experimentation or when the process is not very mature.

GUIDELINES FOR DESIGNING EXPERIMENTS
Choice of experimental design

- Choice of design involves:
  - consideration of sample size (number of replicates),
  - selection of a suitable run order for the experimental trials,
  - and determination of whether blocking or other randomization restrictions are involved.
- Design selection also involves thinking about and selecting a tentative **empirical model** to describe the results.
  - The model is just a quantitative relationship (equation) between the response and the important design factors
  - For example, a **first-order** model in two variables is:
  $$y = \beta_0 + \beta_1 x_1 + \beta_1 x_1 + \varepsilon$$
  where y is the response, x_1 and x_2 are the design factors, β_0, β_1 and β_2 are unknown parameters that will be estimated from the data in the experiment, and ε is a random error term that accounts for the experimental error in the system that is being studied.

GUIDELINES FOR DESIGNING EXPERIMENTS

Performing the experiment

- When running the experiment, it is vital to monitor the process carefully to ensure that everything is being done according to plan.
- Errors in experimental procedure at this stage will usually destroy experimental validity.
- One of the most common mistakes is that the people conducting the experiment failed to set the variables to the proper levels on some runs.
- Coleman and Montgomery (1993) suggest that prior to conducting the experiment a few trial runs or pilot runs are often helpful.
  - These runs provide information about consistency of experimental material, a check on the measurement system, a rough idea of experimental error, and a chance to practice the overall experimental technique.

GUIDELINES FOR DESIGNING EXPERIMENTS

Statistical analysis of the data

- Statistical methods should be used to analyze the data so that results and conclusions are **objective** rather than judgmental in nature.
- Often, we find that simple **graphical methods** play an important role in data analysis and interpretation.
- Because many of the questions that the experimenter wants to answer can be cast into a hypothesis-testing framework, **hypothesis testing** and **confidence interval** estimation procedures are very useful in analyzing data from a designed experiment.
- Residual analysis and model adequacy checking are also important analysis techniques.

GUIDELINES FOR DESIGNING EXPERIMENTS

Statistical analysis of the data

- Remember that statistical methods cannot prove that a factor (or factors) has a particular effect.
- They only provide guidelines as to the reliability and validity of results.
- When properly applied, statistical methods do not allow anything to be proved experimentally, but they do allow us to measure the likely error in a conclusion or to attach a level of confidence to a statement.
- The primary advantage of statistical methods is that they add **objectivity** to the decision-making process.

GUIDELINES FOR DESIGNING EXPERIMENTS

Conclusions and recommendations

- Once the data have been analyzed, the experimenter must draw *practical* conclusions about the results and recommend a course of action.
- Graphical methods are often useful in this stage, particularly in presenting the results to others.
- **Follow-up runs** and **confirmation testing** should also be performed to validate the conclusions from the experiment.

2. SUMMARIZING MEASURED DATA

Summarizing Measured Data

- **Central Tendency**:
  - mean,
  - median,
  - mode.
- **Variability**:
  - range
  - variance,
  - standard deviation
- **Distribution**: type of distribution

Central Tendency

- Tries to capture "center" of a distribution of values
- Use this "center" to summarize overall behavior
- Indices of Central Tendency
  - **Sample mean**: Common "average"
  - **Sample median**: ½ of the values are above, ½ below
  - **Mode**: Most common

Sample Mean

- “Sample” implies that values are measured from a random process on discrete random variable X
- Value computed is only an approximation of true mean value of underlying process
- True mean value cannot actually be known
  - Would require infinite number of measurements
- Expected value of $X = E[X]$

$$E[X] = \sum_{i=1}^{n} x_i p_i$$

- x_i : *values measured*
- $p_i = \Pr(X = x_i) = \Pr(we\ measure\ x_i)$

Sample Mean

- Without additional information, assume
  - p_i = constant = $\frac{1}{n}$
  - n = number of measurements
- **Arithmetic mean**
  - Common "average"

$$\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$$

Sample Mean

- Potential Problem with Means
  - Sample mean gives equal weight to all measurements
  - Outliers can have a large influence on the computed mean value
  - Distorts our intuition about the central tendency of the measured values

Median

- Index of central tendency with ½ of the values larger, ½ smaller
- Sort n measurements
- If n is odd
  - Median = middle value
  - Else, median = mean of two middle values
- Reduces skewing effect of outliers on the value of the index

Median

- Measured values: $10, 20, 15, 18, 16$
  - Mean = 15.8
  - Median = 16
- Obtain one more measurement: 200
  - Mean = 46.5
  - Median = $\frac{1}{2}\ (16\ +\ 18)\ =\ 17$
- Median give more intuitive sense of central tendency

Mode

- Value that occurs most often
- May not exist
- May not be unique
  - E.g. “bi-modal” distribution
    - Two values occur with same frequency

Mean, Median, or Mode?

- Mean
  - If the sum of all values is meaningful
  - Incorporates all available information
- Median
  - Intuitive sense of central tendency with outliers
  - What is “typical” of a set of values?
- Mode
  - When data can be grouped into distinct types, categories (categorical data)

Quantifying variability

- Means hide information about variability
- How “spread out” are the values?
- How much spread relative to the mean?
- What is the shape of the distribution of values?
- Indices of dispersion
  - Range
  - Variance or standard deviation

Index of Dispersion

- Quantifies how “spread out” measurements are
- Range
  - (max value) – (min value)
- Maximum distance from the mean
  - Max of $|x_i - mean|$
- Neither efficiently incorporates all available information

Index of Dispersion

- Sample Variance

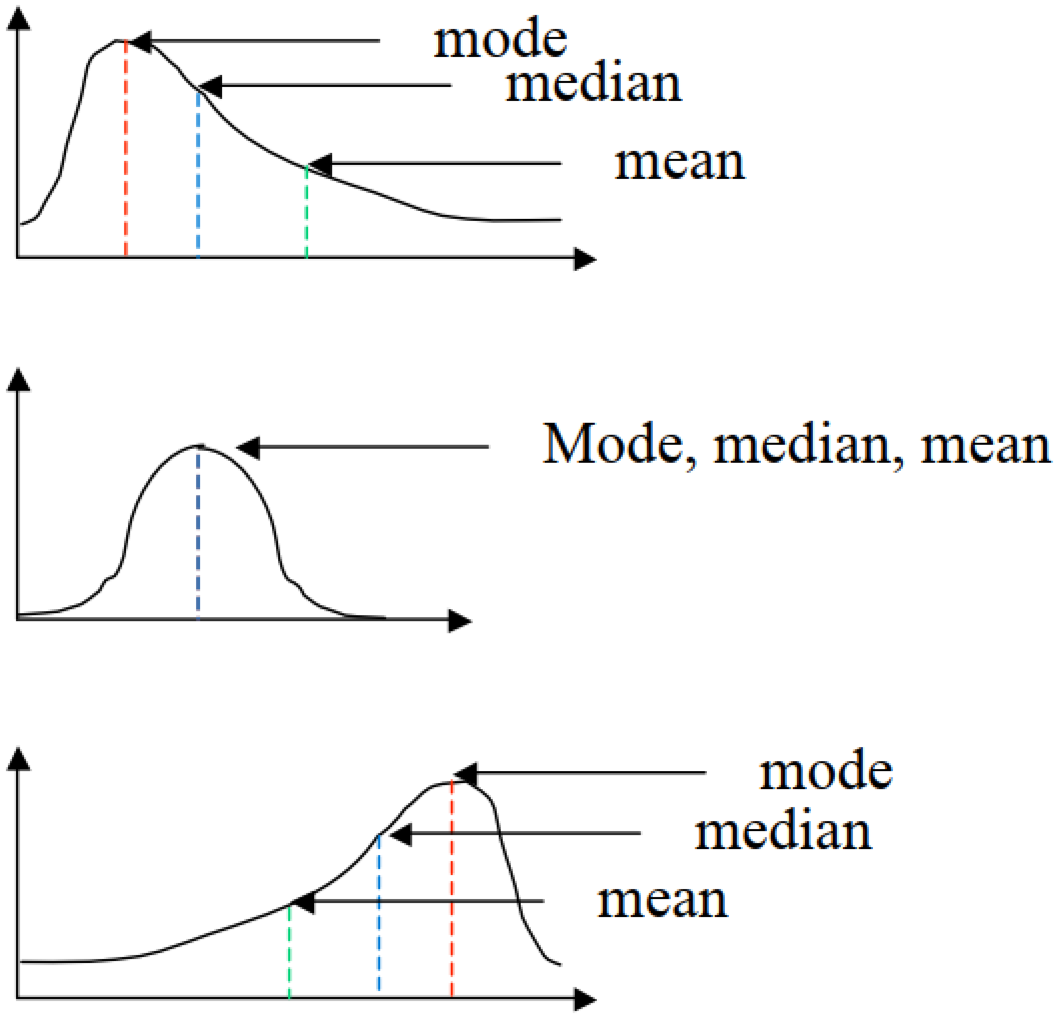
$$s^2 = \frac{\sum_{i=1}^{n}(x_i - \bar{x})^2}{n-1} = \frac{\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{x}_{\mathrm{i}}^2 - \frac{\left(\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{x}_{\mathrm{i}}\right)^2}{\mathrm{n}}}{\mathrm{n}-1}$$

  - Second moment of random variable X
  - Second form good for calculating “on-the-fly”
    - One pass through data
  - (n-1) degrees of freedom

Index of Dispersion

- **Sample Variance**
  - Gives “units-squared”
  - Hard to compare to mean
- Use **standard deviation**, s
  - s = square root of variance
  - Units = same as mean

Shapes of Distributions



Right-skewed distribution

Symmetric distribution

Left-skewed distribution

Determining Distributions

- A measured data set can be summarized by stating its average and variability
- If we can say something about the distribution of the data, that would provide all the information about the data
  - Distribution information is required if the summarized mean and variability have to be used in simulations or analytical models
- To determine the distribution of a data set, we compare the data set to a theoretical distribution
  - Heuristic techniques (Graphical/Visual): Histograms, Q-Q plots
  - Statistical goodness-of-fit tests: Chi-square test, Kolmogrov-Smirnov test

3. MEASUREMENTAL ERRORS and STATISTICS

What are statistics?

- "A branch of mathematics dealing with the collection, analysis, interpretation, and presentation of masses of numerical data."

  Merriam-Webster

- We are most interested in analysis and interpretation here.

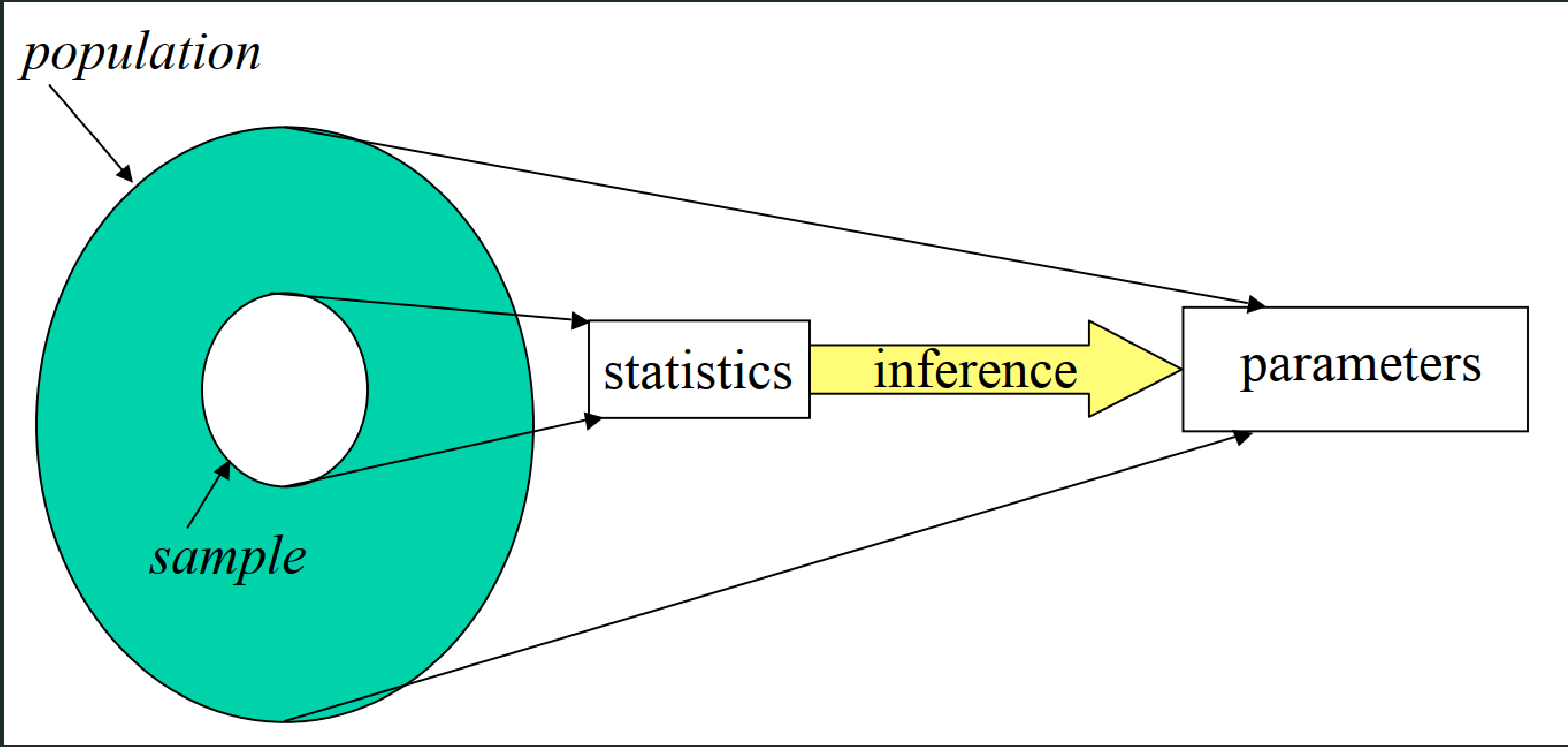
What is a statistic?

- "A quantity that is computed from a sample [of data]."

  Merriam-Webster

- An estimate of a population parameter.

STATISTICAL INFERENCE



Why do we need statistics?

- A set of experimental measurements constitute a sample of the underlying process/system being measured
  - Use statistical techniques to infer the true value of the metric
- Use statistical techniques to quantify the amount of imprecision due to **random experimental errors**

Experimental errors

- Errors → noise in measured values
- Systematic errors
  - Result of an experimental “mistake”
  - Typically produce constant or slowly varying bias
- Controlled through skill of experimenter
- Examples
  - Temperature change causes clock drift
  - Forget to clear cache before timing run

Experimental errors

- **Random errors**
  - Unpredictable, non-deterministic
  - Unbiased → equal probability of increasing or decreasing measured value
- Result of
  - Limitations of measuring tool
  - Observer reading output of tool
  - Random processes within system
- Typically cannot be controlled
  - Use statistical tools to characterize and quantify

Model of Random Errors

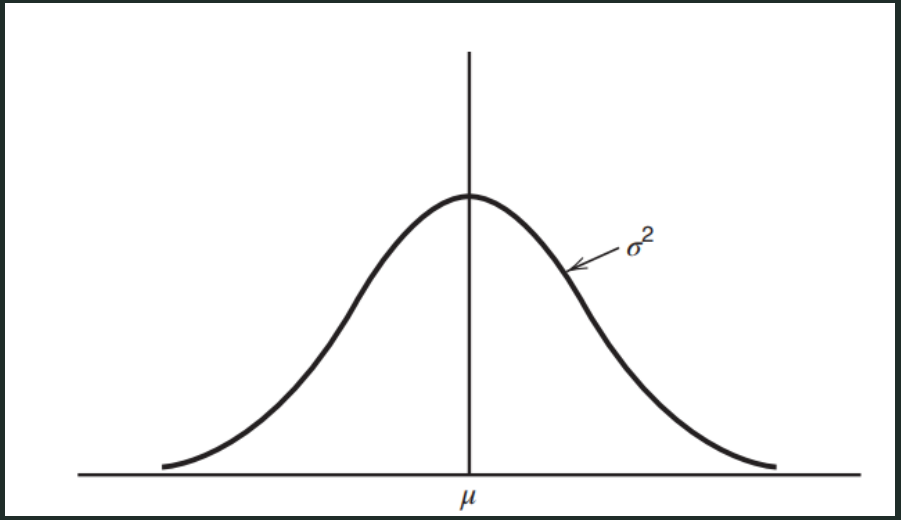
- We assume that random errors are normally and independently distributed random variables with mean 0 and variance σ^2.

The normal distribution

$$f(y) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$$

$-\infty < \mu < \infty$: is the mean of the distribution

$\sigma^2 > 0$: is the variance.



Properties of Point Estimators

- In statistics, **point estimation** involves the use of sample data to calculate a single value which is to serve as a "best guess" for an unknown (fixed or random) population parameter.
- Example of point estimator: sample mean.
- Properties:
  - Unbiasedness: the expected value of all possible sample statistics (of given size n) is equal to the population parameter.

$$E[\bar{X}] = \mu$$
$$E[s^2] = \sigma^2$$

  - Consistency: as the sample size increases the sample statistic becomes a better estimator of the population parameter.

Central Limit Theorem

- If the observations in a sample are independent and come from the same population that has mean μ and standard deviation σ then the sample mean for **large** samples has a normal distribution with mean μ and standard deviation $\frac{\sigma}{\sqrt{n}}$

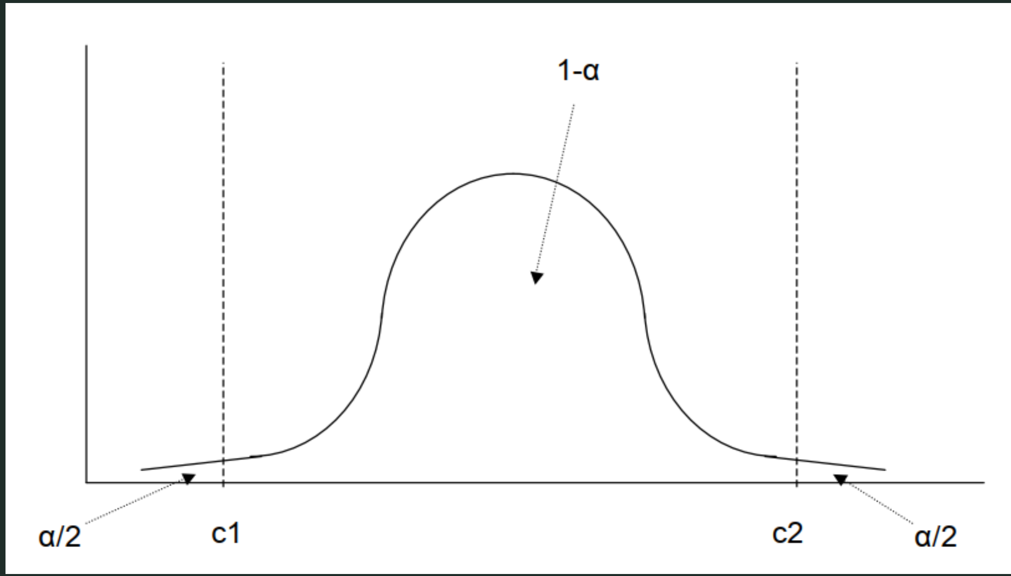
$$\bar{x} \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right)$$

- The standard deviation of the sample mean is called the standard error.

Confidence Intervals

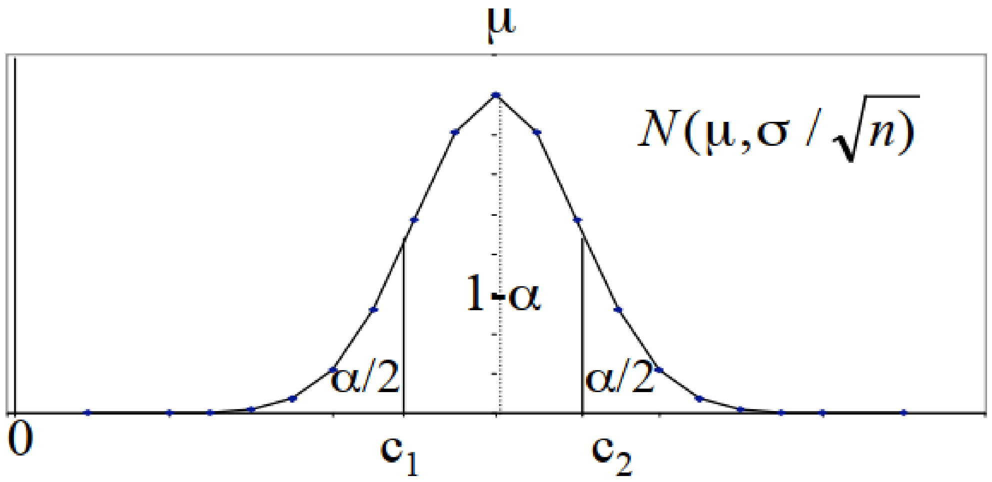
- Confidence intervals quantify the degree of uncertainty associated with the estimation of population parameters such as the mean or the variance.

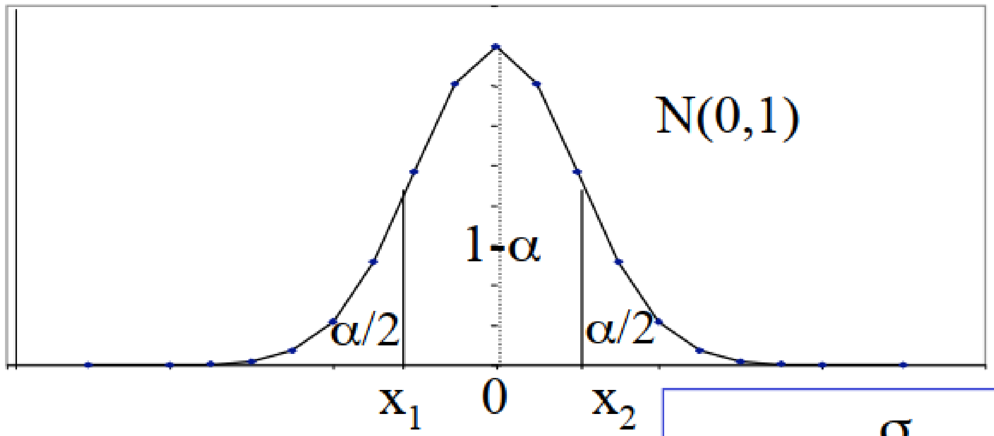
- Can be defined as "the interval that contains the true value of a given population parameter with a confidence level of $100(1 - \alpha)$";



Confidence Intervals for the Mean

$c_2 = Q(1-\alpha/2)$
$c_1 = Q(\alpha/2)$





$x_2 = z_{1-\alpha/2}$
$x_1 = z_{\alpha/2} = -z_{1-\alpha/2}$

$$c_2 = \mu + \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2}$$

$$c_1 = \mu + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} = \mu - \frac{\sigma}{\sqrt{n}} z_{1-\alpha/2}$$

Confidence Intervals for the Mean
Large ($n > 30$) Samples

- $100\,(1-\alpha)\%$ confidence interval for the population mean:

$$\left(\bar{x} - \mathrm{z}_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}, \bar{x} + \mathrm{z}_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}}\right)$$

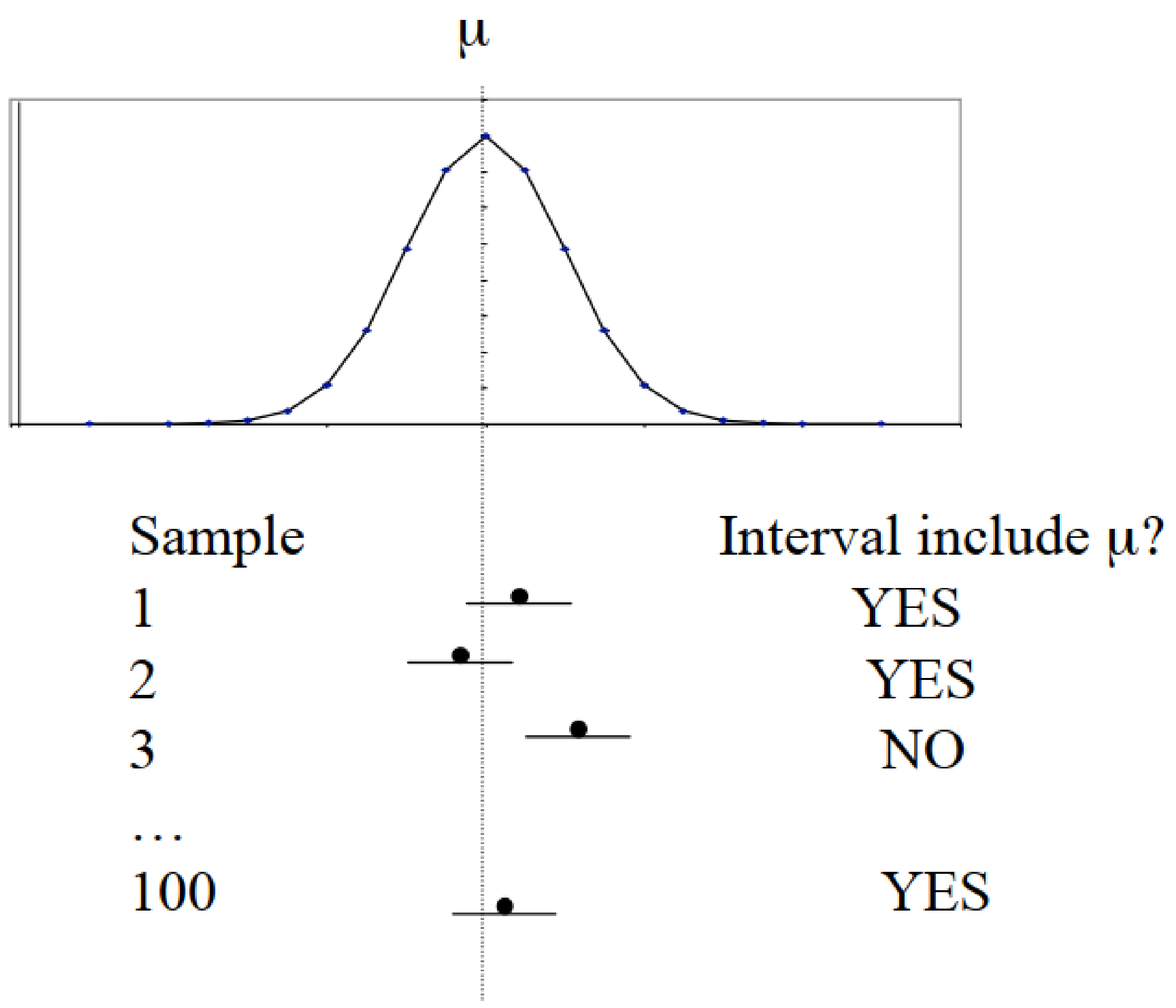
$\bar{x}$: sample mean

s: sample standard deviation

n: sample size

$\mathrm{z}_{1-\frac{\alpha}{2}}$: $\left(1-\frac{\alpha}{2}\right)$-quantile of a unit normal variate $\left(\mathrm{N}(0,1)\right)$

Confidence Intervals for the Mean



100 (1 – α) of the 100 samples include the population mean μ.

How many measurements do we need for a desired interval width?

- Width of interval inversely proportional to $\sqrt{n}$
- Want to minimize number of measurements
- Find confidence interval for mean, such that:
  - Pr(actual mean in interval) = (1 – α)

$$(c_1, c_2) = [(1-e)\bar{x}, (1+e)\bar{x}]$$

How many measurements do we need for a desired interval width?

$$(c_1, c_2) = (1 \pm e)\bar{x} = \bar{x} \pm z_{1-\alpha/2} \cdot \frac{s}{\sqrt{n}}$$

$$\Longrightarrow z_{1-\frac{\alpha}{2}} \cdot \frac{s}{\sqrt{n}} = \bar{x}e$$

$$\Longrightarrow n = \left(\frac{z_{1-\alpha/2}S}{\bar{x}e}\right)^2$$

How many measurements do we need for a desired interval width?

- But n depends on knowing mean and standard deviation!
- Estimate s with small number of measurements
- Use this s to find n needed for desired interval width

How many measurements do we need for a desired interval width?

- Mean = 7.94 s
- Standard deviation = 2.14 s
- Want 90% confidence mean is within 7% of actual mean.

How many measurements do we need for a desired interval width?

- Mean = 7.94 s
- Standard deviation = 2.14 s
- Want 90% confidence mean is within 7% of actual mean.

- $\alpha = 0.90$
- $(1-\alpha/2) = 0.95$
- Error = ± 3.5%
- $e = 0.035$

How many measurements do we need for a desired interval width?

$$n = \left(\frac{z_{1-\alpha/2} S}{\bar{x} e}\right)^2 = \left(\frac{1.895 \cdot 2.14}{0.035 \cdot 7.94}\right)^2 = 212.9$$

- 213 measurements

  → 90% chance true mean is within ± 3.5% interval

REFERENCES

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